

# Brent Minchew

Professor of Geophysics  
Division of Geological and Planetary Sciences  
California Institute of Technology

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1200 E. California Blvd.  
Pasadena, CA 91125

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## Curriculum Vitae

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### Bio

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I study how glaciers and ice sheets respond to climate change, with emphasis on the fundamental mechanisms that govern rates of sea-level rise in my daughter's lifetime. My research leverages satellite remote sensing, mathematical theory, and computational models to study the flow, deformation, and fracture of glacier ice across scales from atomic to continental and seconds to centuries. My work allows me to see nature at its most elegant, both fragile and resilient, stately and responsive.

### Education

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2016 Ph.D. Geophysics, California Institute of Technology  
2010 M.S. Aerospace Engineering, University of Texas at Austin  
2008 B.S. Aerospace Engineering, University of Texas at Austin

### Academic Positions

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|-------------|------------------------------------------------------------------------------|
| 2025–       | Professor of Geophysics, California Institute of Technology                  |
| 2024–2025   | Chair of the Program in Geophysics, Massachusetts Institute of Technology    |
| 2024–2025   | Associate Professor (with tenure), Massachusetts Institute of Technology     |
| 2023–2024   | Associate Professor (without tenure), Massachusetts Institute of Technology  |
| 2018–2023   | Assistant Professor, Massachusetts Institute of Technology                   |
| 2016–2018   | NSF Postdoctoral Fellow, British Antarctic Survey                            |
| 2010–2015   | Graduate Research and Teaching Assistant, California Institute of Technology |
| 2009 & 2010 | Graduate Research Fellow, NASA Jet Propulsion Laboratory                     |
| 2006–2010   | Research and Teaching Assistant, University of Texas at Austin               |

### Nonacademic Positions

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|-----------|----------------------------------------------------------|
| 2024–     | Co-Founder and Chief Scientist, Arête Glacier Initiative |
| 1996–2004 | Staff Sergeant, United States Marine Corps               |

### Select Awards and Honors

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|------|-----------------------------------------------------|
| 2022 | Class of 1948 Career Development Professorship, MIT |
| 2019 | Cecil and Ida Green Career Development Chair, MIT   |

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| 2014      | NSF Earth Sciences Postdoctoral Fellowship                               |
| 2013–2015 | ARCS and Albert Parvin Foundation Fellowship                             |
| 2012      | IEEE Transactions on Geoscience and Remote Sensing Editor's Choice Award |
| 2011–2014 | NASA Earth and Space Sciences Fellowship                                 |
| 2011–2013 | ARCS Foundation Fellowship                                               |

## Publications and Presentations

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### Scholarly articles

(46 in total; \* indicates students and \* postdocs and other group members)

46. S. Wells-Moran, **B. M. Minchew**, and B. V. Riel. “Near-total loss of buttressing stresses observed on Pine Island Ice Shelf, West Antarctica”. *Proceedings of the National Academy of Sciences* 123.33 (2026), e2602994123. DOI: 10.1073/pnas.2602994123.
45. C. R. Meyer, K. L. P. Warburton, A. N. Sommers, and **B. M. Minchew**. “A model of water extraction from the subglacial hydrologic system under idealized conditions”. *The Cryosphere* 20.5 (2026), pp. 2659–2680. DOI: 10.5194/tc-20-2659-2026.
44. D. F. Martin, S. B. Kachuck, J. D. Millstein, and **B. M. Minchew**. “Rates of sea-level rise are highly sensitive to ice viscosity parameters in model benchmarks”. *AGU Advances* 7.2 (2026), e2025AV001946. DOI: 10.1029/2025AV001946.
43. F. M. Elgart\*, **B. M. Minchew**, and C. R. Meyer. “Estimating effective elasticity in grounding zones of the Ross Ice Shelf with tidal flexure from ICESat-2”. *Journal of Glaciology* 72 (2026), e23. DOI: 10.1017/jog.2026.10137.
42. C. R. Meyer, A. G. Stubblefield, **B. M. Minchew**, S. S. Pegler, A. Berne, J. J. Buffo, and T. C. Tomlinson. “Implications of shallow-shell models for topographic relaxation on icy satellites”. *Journal of Geophysical Research - Planets* 130.12 (2025), e2025JE009247. DOI: 10.1029/2025JE009247.
41. N. Narayanan\*, A. Sommer, W. Chu, J. Steiner, M. A. Siddique, C. R. Meyer, and **B. M. Minchew**. “Simulating seasonal evolution of subglacial hydrology at a surging glacier in the Karakoram”. *Journal of Glaciology* 71 (2025), e94. DOI: 10.1017/jog.2025.10078.
40. S. Wells-Moran\*\*, M. Ranganathan\*, and **B. M. Minchew**. “Fracture criteria and tensile strength for natural glacier ice calibrated from remote sensing observations of Antarctic ice shelves”. *Journal of Glaciology* 71 (2025), e47. DOI: 10.1017/jog.2024.104.
39. U. Ovienmhada\*, M. Hines-Shanks, M. Krisch, A. Diongue, **B. M. Minchew**, and D. Wood. “Spatiotemporal Facility-Level Patterns of Summer Heat Exposure, Vulnerability, and Risk in United States Prison landscapes”. *GeoHealth* 8.9 (2024). DOI: 10.1029/2024GH001108.
38. M. Ranganathan\* and **B. M. Minchew**. “A modified viscous flow law for natural glacier ice: Scaling from laboratories to ice sheets”. *Proceedings of National Academy of Sciences* 121.23 (2024). DOI: 10.1073/pnas.2309788121.

37. F. Clerc\*, M. D. Behn, and **B. M. Minchew**. “Deglaciation-enhanced mantle CO<sub>2</sub> fluxes at Yellowstone imply positive climate feedbacks”. *Nature Communications* 15.1526 (2024), pp. 1–10. DOI: 10.1038/s41467-024-45890-z.
36. Y. Sun\*, B. V. Riel\*, and **B. M. Minchew**. “Disintegration and Buttressing Effect of the Landfast Sea Ice in the Larsen B Embayment, Antarctic Peninsula”. *Geophysical Research Letters* 50.16 (2023). DOI: 10.1029/2023GL104066.
35. M. Ranganathan\*, J. W. Barotta\*\*, C. R. Meyer, and **B. M. Minchew**. “Meltwater generation in ice stream shear margins: Case study in Antarctic ice streams”. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 479.2273 (2023). DOI: 10.1098/rspa.2022.0473.
34. B. V. Riel\* and **B. M. Minchew**. “Variational inference of ice shelf rheology with physics-informed machine learning”. *Journal of Glaciology* 69.277 (2023), pp. 1167–1186. DOI: 10.1017/jog.2023.8.
33. M. Zhong, M. Simons, **B. M. Minchew**, and L. Zhu. “Inferring tide-induced ephemeral grounding and subsequent dynamical response in an ice-shelf-stream system: Rutford Ice Stream, West Antarctica”. *Journal of Geophysical Research – Earth Surface* 128.2 (2023). DOI: 10.1029/2022JF006789.
32. L. Ultee\*, D. Felikson, **B. M. Minchew**, L. A. Stearns, and B. V. Riel\*. “Helheim Glacier ice velocity variability responds to runoff and terminus position change at different timescales”. *Nature Communications* 13.6022 (2022). DOI: 10.1038/s41467-022-33292-y.
31. J. D. Millstein\*, **B. M. Minchew**, and S. S. Pegler. “Ice viscosity is more sensitive to stress than commonly assumed”. *Nature Communications Earth and Environment* 3.57 (2022). DOI: 10.1038/s43247-022-00385-x.
30. M. Ranganathan\*, **B. M. Minchew**, C. R. Meyer, and M. Pec. “Recrystallization of ice enhances creep and the vulnerability to fracture of ice shelves”. *Earth and Planetary Science Letters* 576 (2021). DOI: 10.1016/j.epsl.2021.117219.
29. B. V. Riel\*, **B. M. Minchew**, and T. Bischoff. “Data-driven inference of the mechanics of slip along glacier beds using physics-informed neural networks: Case study on Rutford Ice Stream, Antarctica”. *Journal of Advances in Modeling Earth Systems* 13.11 (2021), e2021MS002621. DOI: 10.1029/2021MS002621.
28. G. Guerin\*\*, A. Mordret\*, D. Rivet, B. P. Lipovsky, and **B. M. Minchew**. “Frictional origin of slip events of the Whillans Ice Stream, Antarctica”. *Geophysical Research Letters* 48.11 (2021), pp. 1–9. DOI: 10.1029/2021GL092950.
27. K. S. Shah\*, S. S. Pegler, and **B. M. Minchew**. “Dynamics of two-layer fluid flows on inclined surfaces”. *Journal of Fluid Mechanics* 917.A54 (2021), pp. 1–33. DOI: 10.1017/jfm.2021.273.
26. P. Hunter\*, C. R. Meyer, **B. M. Minchew**, M. Haseloff, and A. Rempel. “Thermal controls on ice stream shear margins”. *Journal of Glaciology* 67.263 (2021), pp. 435–449. DOI: 10.1017/jog.2020.118.
25. B. V. Riel\*, **B. M. Minchew**, and I. Joughin. “Observing traveling waves in glaciers with remote sensing: New flexible time-series methods and application to Sermeq Kujalleq (Jakobshavn Isbræ), Greenland”. *The Cryosphere* 15.1 (2021), pp. 407–429. DOI: 10.5194/tc-15-407-2021.

24. M. Ranganathan\*, **B. M. Minchew**, C. R. Meyer, and G. H. Gudmundsson. “A new approach to inferring basal drag and ice rheology in ice streams, with applications to West Antarctic ice streams”. *Journal of Glaciology* 67.262 (2021). DOI: 10.1017/jog.2020.95.
23. E. H. Ultee\*, C. R. Meyer, and **B. M. Minchew**. “Tensile strength of glacial ice deduced from observations of the 2015 Eastern Skaftá Cauldron collapse, Vatnajökull ice cap, Iceland”. *Journal of Glaciology* 66.260 (2020), pp. 1024–1033. DOI: 10.1017/jog.2020.65.
22. **B. M. Minchew** and C. R. Meyer. “Dilation of subglacial sediment governs incipient surge motion in glaciers with deformable beds”. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 476.2238 (2020). DOI: 10.1098/rspa.2020.0033.
21. **B. M. Minchew** and I. Joughin. “Toward a universal glacier slip law”. *Science* 368.6486 (2020), pp. 29–30. DOI: 10.1126/science.abb3566.
20. F. Clerc\*, **B. M. Minchew**, and M. D. Behn. “Marine ice cliff instability mitigated by slow removal of ice shelves”. *Geophysical Research Letters* 46.21 (2019), pp. 12108–12116. DOI: 10.1029/2019GL084183.
19. **B. M. Minchew**, C. R. Meyer, S. S. Pegler, B. P. Lipovsky, A. W. Rempel, G. H. Gudmundsson, and N. R. Iverson. “Comment on: “Friction at the bed does not control fast glacier flow” by L. A. Stearns and C. J. van der Veen”. *Science* 363.6427 (2019). DOI: 10.1126/science.aau6055.
18. C. R. Meyer, A. Yehya, **B. M. Minchew**, and J. R. Rice. “A model for the downstream evolution of temperate ice and subglacial hydrology along ice stream shear margins”. *Journal of Geophysical Research - Earth Surface* 123.8 (2018), pp. 1682–1698. DOI: 10.1029/2018JF004669.
17. C. R. Meyer and **B. M. Minchew**. “Temperate ice in the shear margins of the Antarctic Ice Sheet: controlling processes and preliminary locations”. *Earth and Planetary Science Letters* 498 (2018), pp. 17–26. DOI: 10.1016/j.epsl.2018.06.028.
16. **B. M. Minchew**, C. R. Meyer, A. A. Robel, G. H. Gudmundsson, and M. Simons. “Processes controlling the downstream evolution of ice rheology in glacier shear margins: Case study on Rutford Ice Stream, West Antarctica”. *Journal of Glaciology* 64.246 (2018), pp. 583–594. DOI: 10.1017/jog.2018.47.
15. **B. M. Minchew**, G. H. Gudmundsson, A. Gardner, F. S. Paolo, and H. A. Fricker. “Modeling the dynamic response of outlet glaciers to observed ice-shelf thinning in the Bellingshausen Sea Sector, West Antarctica”. *Journal of Glaciology* 64.244 (2018), pp. 333–342. DOI: 10.1017/jog.2018.24.
14. S. Angelliaume, P. Dubois-Fernandez, C. E. Jones, B. Holt, **B. M. Minchew**, E. Amri, and V. Mieggebielle. “SAR imagery for detecting sea surface slicks: Performance assessment of polarimetric parameters”. *IEEE Transactions on Geoscience and Remote Sensing* 56.8 (2018), pp. 4237–4257. DOI: 10.1109/TGRS.2018.2803216.
13. A. A. Robel, V. C. Tsai, **B. M. Minchew**, and M. Simons. “Tidal modulation of ice shelf buttressing stresses”. *Annals of Glaciology* 58.74 (2017), pp. 12–20. DOI: 10.1017/aog.2017.22.

12. P. Milillo, **B. M. Minchew**, M. Simons, P. Agram, and B. Riel. “Geodetic imaging of time-dependent three-component surface deformation: application to tidal-timescale ice flow of Rutford Ice Stream, West Antarctica”. *IEEE Transactions on Geoscience and Remote Sensing* 55.10 (2017), pp. 5515–5524. DOI: 10.1109/TGRS.2017.2709783.
11. S. Angelliaume, **B. M. Minchew**, S. Chatiang, P. Martineau, and V. Miegebielle. “Multifrequency radar imagery and characterization of hazardous and noxious substances at sea”. *IEEE Transactions on Geoscience and Remote Sensing* 55.5 (2017). DOI: 10.1109/TGRS.2017.2661325.
10. **B. M. Minchew**, M. Simons, B. V. Riel, and P. Milillo. “Tidally induced variations in vertical and horizontal motion on Rutford Ice Stream, West Antarctica, inferred from remotely sensed observations”. *Journal of Geophysical Research: Earth Surface* 122 (2017), pp. 167–190. DOI: 10.1002/2016JF003971.
9. **B. M. Minchew**, M. Simons, H. Björnsson, F. Pálsson, M. Morlighem, H. Seroussi, E. Larour, and S. Hensley. “Plastic bed beneath Hofsjökull Ice Cap, central Iceland, and the sensitivity of ice flow to surface meltwater flux”. *Journal of Glaciology* 62.231 (2016), pp. 147–158. DOI: 10.1017/jog.2016.26.
8. P. Milillo, B. Riel, **B. M. Minchew**, S. H. Yun, M. Simons, and P. Lundgren. “On the synergistic use of SAR constellations’ data exploitation for earth science and natural hazard response”. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 9.3 (2015), pp. 1095–1100. DOI: 10.1109/JSTARS.2015.2465166.
7. M. J. Collins, M. Denbina, **B. M. Minchew**, C.E. Jones, and B. Holt. “On the use of simulated airborne compact polarimetric SAR for characterizing oil-water mixing of the Deepwater Horizon oil spill”. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 8.3 (2015), pp. 1062–1077. DOI: 10.1109/JSTARS.2015.2401041.
6. **B. M. Minchew**, M. Simons, S. Hensley, H. Björnsson, and F. Pálsson. “Early melt-season velocity fields of Langjökull and Hofsjökull ice caps, central Iceland”. *Journal of Glaciology* 61.226 (2015), pp. 253–266. DOI: 10.3189/2015JG14J023.
5. J. S. Scheingross, **B. M. Minchew**, B.H. Mackey, M. Simons, M.P. Lamb, and S. Hensley. “Fault zone controls on the spatial distribution of slow-moving landslides”. *GSA Bulletin* 125.3-4 (2013), pp. 473–489. DOI: 10.1130/B30719.1.
4. **B. M. Minchew**, C.E. Jones, and B. Holt. “Polarimetric analysis of backscatter from the Deepwater Horizon oil spill using L-band synthetic aperture radar”. *IEEE Transactions on Geoscience and Remote Sensing* 50.10 (2012), pp. 3812–3830. DOI: 10.1109/TGRS.2012.2185804.
3. **B. M. Minchew**. “Determining the mixing of oil and seawater using polarimetric synthetic aperture radar”. *Geophysical Research Letters* 39.16 (2012). L16607. DOI: 10.1029/2012GL052304.
2. V. C. Tsai, **B. M. Minchew**, M. P. Lamb, and J. P. Ampuero. “A physical model for seismic noise generation from sediment transport in rivers”. *Geophysical Research Letters* 39.2 (2012). L02404. DOI: 10.1029/2011GL050255.
1. C. E. Jones, **B. M. Minchew**, B. Holt, and S. Hensley. “Studies of the Deepwater Horizon Oil Spill with the UAVSAR radar”. *Monitoring and Modeling the Deepwater Horizon Oil Spill: A Record-Breaking Enterprise*. Vol. 195. Washington, DC: AGU, 2011, pp. 33–50. DOI: 10.1029/2011GM001113.

## Other publications

2. Brent Minchew and Colin Meyer. “We study glaciers. ‘Artificial glaciers’ and other tech may halt their total collapse”. *The Guardian* (2026). Op-ed, originally entitled: “Sea levels are rising rapidly, threatening the homes of millions. Let’s do something about it”.
1. B. M. Minchew, L. Mahle, P. Ghosh, M. Sistla, and E. N. Martin. “Polar infrastructure and science for national security: A federal agenda to promote glacier resilience and strengthen American competitiveness”. *Day One Project, Federation of American Scientists* (2024).

## Working manuscripts and preprints

(Group members and advisees: \* indicates graduate students, \*\* undergraduates, and \* postdocs and research scientists)

7. B. V. Riel, M. Ranganathan, and **B. M. Minchew**. “Time-varying velocities recover effective ice shelf rheology hidden from snapshots”. *submitted* (2026).
6. E. Kramer E.\* and **B. M. Minchew**. “Space-based InSAR with sub-daily temporal baselines”. *submitted* (2026). DOI: 10.31223/X5XV3J.
5. A. S. Miller\*, J. T. Cole\*\*, C. R. Meyer, and **B. M. Minchew**. “Models of Thermosyphon Performance, Design, and Influence on Glacier Flow”. *submitted* (2026). DOI: 10.31223/X56Z19.
4. B. V. Riel and **B. M. Minchew**. “Identifying the mechanisms behind seasonal velocity changes in Greenland glaciers”. *submitted* (2026).
3. E. Kramer\* and **B. M. Minchew**. “Magnitude-order improvements to the measurement precision of Antarctic grounding zone dynamics using SAR satellite constellations”. *submitted* (2026). DOI: 10.31223/X5RV30.
2. F. M. Elgart\* and **B. M. Minchew**. “Estimating Ice Shelf Thickness in Grounding Zones of the Filchner-Ronne Ice Shelf with Tidal Flexure from ICESat-2”. *in review* (2026).
1. C. Mouchon\*, **B. M. Minchew**, and C. R. Meyer. “Evolution of ice viscosity with stress as constrained by observations of a surge on Vavilov Ice Cap”. *in review* (2026).

## Invited presentations

(past 5 years only)

- B. M. Minchew. “Rheology: The workhorse of glaciology”. *Challenges, Advances, and Emerging Directions in Ice Deformation*. Apr. 2026.
- B. M. Minchew. “Glacier dynamics from viscous flow to iceberg calving”. *UCLA EPSS Colloquium*. Jan. 2026.
- B. M. Minchew. “The rapid demise of a major outlet glacier in West Antarctica”. *Caltech Geo Club*. Dec. 2025.
- B. M. Minchew. “How fast will sea-levels rise and what are we going to do about it?” *Busan IAMAS-IACS-IAPSO Joint Assembly 2025*. 2025.
- B. M. Minchew. “Some ideas for slowing glaciers and sea-level rise”. *UC Santa Barbara Kavli Institute for Theoretical Physics*. 2025.

- B. M. Minchew. “How fast will sea-levels rise and what are we going to do about it?” *MIT EVDT Community Meeting*. May 2025.
- B. M. Minchew. “Glacier dynamics from viscous flow to iceberg calving”. *Geodynamics Seminar, Woods Hole Oceanographic Institution*. 2025.
- B. M. Minchew. “Glacier dynamics from viscous flow to iceberg calving”. *Geodynamics Seminar, Lamont Doherty Earth Observatory, Columbia University*. 2025.
- B. M. Minchew and C. R. Meyer. “Between a rock and hard place: Some thoughts on how to approach responsible glacier interventions”. *AGU Fall Meeting*. Dec. 2024.
- B. M. Minchew. “Slowing glaciers by increasing drag at the bed: A brief primer”. *Glacial Geo-engineering Workshop, Stanford University*. Dec. 2023.
- B. M. Minchew. “Glacier dynamics from viscous flow to iceberg calving”. *Geological and Planetary Sciences Division Seminar, California Institute of Technology*. Nov. 2023.
- B. M. Minchew. “Principles behind modulating basal slip in fast-flowing glaciers”. *Glacial Geo-engineering Workshop, University of Chicago*. Oct. 2023.
- B. M. Minchew. “Our evolving understanding of the flow, deformation, and fracture of glacier ice”. *Geology and Geophysics Department Seminar, Woods Hole Oceanographic Institute*. Aug. 2023.
- B. M. Minchew. “Our evolving understanding of the flow, deformation, and fracture of glacier ice”. *Geophysics Seminar, Stanford University*. Apr. 2023.
- B. M. Minchew. “Our evolving understanding of the flow, deformation, and fracture of glacier ice”. *Department of Earth, Environmental, and Planetary Sciences Seminar, Brown University*. Feb. 2023.
- B. M. Minchew. “Our evolving understanding of the flow, deformation, and fracture of glacier ice”. *Earth and Environmental Science Seminar, University of Pennsylvania*. Feb. 2023.
- B. M. Minchew. “Our evolving understanding of the flow, deformation, and fracture of glacier ice”. *University of California, Santa Barbara*. Feb. 2023.
- B. M. Minchew. “Observing at currently unobservable space and time scales”. *Caltech Seismo Lab Centennial*. Nov. 2022.
- B. M. Minchew. “Our evolving understanding of the deformation of glacier ice”. *Tufts University, Civil and Environmental Engineering Colloquium*. Oct. 2022.
- B. M. Minchew. “Sea-level rise in the 21st century: Education and innovation driving science and adaptation”. *MIT-Portugal Annual Conference*. Sept. 2022.
- B. M. Minchew. “Earth Science in the age of NISAR”. *NISAR Science Community Workshop*. Aug. 2022.
- B. M. Minchew. “Managing the risks of sea-level rise through improved ice-sheet monitoring”. *Eureka meets the Atlantic | Boston*. Apr. 2022.
- B. M. Minchew. “Flow and deformation of glacier ice”. *Seismolab Seminar, California Institute of Technology*. Mar. 2022.
- B. M. Minchew. “Lessons learned from observing Earth’s cryosphere from satellites”. *Lowell Observatory, I Heart Pluto Festival*. Feb. 2022.

## Grants

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| 2024–     | PI – Grantham Foundation: Reducing the risk of sea-level rise by deliberately slowing glaciers in Antarctica: First steps.                                                                                                      |
| 2021–     | PI – John W. Jarve (1978) Seed Fund: Improving projections of sea-level rise by (deep) learning glacier dynamics from data. With Bryan Riel (MIT)                                                                               |
| 2021      | PI – MIT Climate Grand Challenges: Ice Sheet Stability and Rapid Sea-Level Rise: Exploring Risk, Impact, and Mitigation Strategies                                                                                              |
| 2021      | Researcher – MIT Climate Grand Challenges: Stratospheric Airborne Climate Observatory System to Initiate Revolution In Climate Risk Forecasting. With PI John Hansman (MIT)                                                     |
| 2021–2022 | Researcher – National Geographic Society: AI for Glacial Lake Outburst Floods Hazard Potential Assessment in Chitral, Pakistan (AI4GLOF). With PI Muhammad Adnan Siddique (Information Technology University, Lahore, Pakistan) |
| 2020–2021 | PI – JPL Strategic University Research Partnership: Ice sheet mechanical properties as revealed from time-varying surface velocity fields. Award number 1657974.                                                                |
| 2020–2022 | PI – NEC Corporation Fund for Research in Computers and Communications: Quantifying the Evolution of Stress Fields in Antarctic Ice Shelves by Fusing Data from Multiple Remote Sensing Platforms, Instruments, and Techniques. |
| 2020–2023 | PI – NSFGE0-NERC: Collaborative Research: A new mechanistic framework for modeling rift processes in Antarctic ice shelves validated through improved strain-rate and seismic observations. Award number 1853918                |
| 2020–2021 | PI – Microsoft AI for Earth: Theory-guided discovery of glacier dynamics using deep learning. With Bryan Riel (MIT)                                                                                                             |
| 2020–2021 | PI – MIT-Germany - FAU Seed Fund: Rift Formation in Antarctic Ice Shelves: Constraining the Mechanical Properties of Glacier Ice Using Time-Dependent, High-Resolution Strain-Rate Fields. With Matthias Braun (FAU)            |
| 2019–2020 | PI – Earl A Killian III (1978) and Waidy Lee Fund: Improving projections of sea-level rise by (deep) learning glacier dynamics from data. With Bryan Riel (MIT)                                                                 |
| 2019–2020 | Co-I – Microsoft AI for Earth: Theory-guided discovery of glacier dynamics using deep learning. With Bryan Riel (MIT) and Tobias Bischoff (Microsoft)                                                                           |
| 2018–2023 | Researcher – NSF-NERC: Processes, drivers, predictions: Modeling the history and evolution of Thwaites Glacier (PROPHET). Award number 1739031                                                                                  |
| 2016–2018 | PI – Spatiotemporal characteristics of basal resistance to ice flow in the West Antarctic Ice Sheet from satellite observations and numerical modeling. NSF Earth Sciences Postdoctoral Fellowship award 1452587                |
| 2013–2015 | Research assistant – Subglacial mechanics using repeat-pass InSAR measurements and numerical models of temperate ice caps in Iceland. NASA Cryospheric Science award NNX14AH80G, with PI Mark Simons (Caltech)                  |
| 2011–2014 | PI – Investigating the mechanics of subglacial till using airborne radar interferometry and numerical ice flow models. NASA Earth and Space Science Fellowship                                                                  |
| 2011–2012 | Research assistant – Temperate glacier studies with UAVSAR. NASA Cryospheric Science, with PI Mark Simons (Caltech)                                                                                                             |

## Mentorship

### Postdoctoral researchers

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|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026–     | Nicolas Morales Preciado (University of British Columbia PhD ‘26), starting 2026                                                                |
| 2026–     | Natasha Morgan-Witts (University of Wisconsin, Madison PhD ‘26), starting July 2026                                                             |
| 2026–     | George Lu (Columbia University PhD ‘26), starting October 2026                                                                                  |
| 2025–     | Evan Kramer (MIT PhD ‘25), Caltech Seismo Lab Director’s Fellow                                                                                 |
| 2018–2021 | Elizabeth (Lizz) Ultee (University of Michigan PhD ‘18); now research scientist at NASA Goddard Space Flight Center and Morgan State University |

### Graduate students, as a primary advisor

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|-----------|-------------------------------------------------------------------------------------------------------------------|
| 2024–     | Thatcher Chamberlin (PhD, MIT EAPS (PAOC))                                                                        |
| 2023–     | Max Filter (PhD, MIT EAPS (Geophysics))                                                                           |
| 2023–2025 | Sarah Wells-Moran (MIT MS ‘25), now PhD student at University of Chicago                                          |
| 2022–2024 | Ufuoma Oviemhada (MIT PhD ‘24)                                                                                    |
| 2021–2023 | Neosha Narayanan (MIT MS ‘23), now PhD student at Georgia Tech                                                    |
| 2019–2025 | Faye Hendley Elgart (MIT PhD ‘25), now postdoc at Laboratoire de Glaciologie at the Université libre de Bruxelles |
| 2019–     | Justin Linick (PhD, MIT EAPS (Geophysics))                                                                        |
| 2018–2023 | Joanna Millstein (MIT-WHOI JP PhD ‘23), now postdoc at Colorado School of Mines                                   |
| 2018–2022 | Fiona Clerc (MIT-WHOI JP PhD ‘22), now faculty at Stony Brook University                                          |
| 2018–2022 | Meghana Ranganathan (MIT PhD ‘22), now faculty at University of Chicago                                           |
| 2018–2019 | Brindha Kanniah (MIT MS ‘19)                                                                                      |

### Graduate students, as a secondary or project advisor

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| 2025–     | PhD project advisor: Jackson Fellows, Caltech GPS (Seismo Lab)                                          |
| 2025–     | PhD project advisor: Mma Ikwut-Ukwa, Caltech GPS (Planetary)                                            |
| 2025–     | PhD project advisor: Ben Reynolds, Caltech GPS (Geology)                                                |
| 2025–     | PhD project advisor: Elena Dworak, Caltech GPS (ESE)                                                    |
| 2025–     | PhD project advisor: Yi (Lain) Li, Caltech GPS (Seismo Lab)                                             |
| 2024–     | PhD project advisor: Caroline Needell, MIT-WHOI Joint Program                                           |
| 2023–     | PhD project advisor: Alex Miller, MIT AeroAstro                                                         |
| 2022–     | PhD project advisor: Caroline Mouchon, MIT EAPS (Geophysics)                                            |
| 2022–2024 | PhD project advisor: Annick Dewald (MIT PhD ‘24, AeroAstro), now founding faculty at Greenway Institute |
| 2021–2023 | PhD project advisor: Grace O’Neil, EAPS (PAOC)                                                          |
| 2021–2023 | PhD project advisor: Yudong Sun (MIT PhD ‘25), now postdoc at Stanford University                       |
| 2021–2023 | PhD project advisor: Mila Lubeck, EAPS (Geophysics)                                                     |
| 2019–2020 | MS project co-advisor: Pierce Hunter (UO MS ‘20)                                                        |

|           |                                                                                                                      |
|-----------|----------------------------------------------------------------------------------------------------------------------|
| 2019–2024 | PhD project advisor: Erik Tamre (MIT PhD ‘24), now postdoc at Dartmouth College                                      |
| 2018–2022 | PhD project advisor: Kasturi Shah (MIT PhD ‘22), now McDonnell Foundation Research Fellow at University of Cambridge |
| 2018–2019 | PhD project advisor: Julia Wilcots (MIT PhD ‘22), now faculty at University of Minnesota, Twin Cities                |

### Undergraduate students

(UROP = MIT Undergrad Research Opportunities Program; MSRP = MIT Summer Research Program)

|           |                                                               |
|-----------|---------------------------------------------------------------|
| 2024      | UROP advisor: Justin Cole, MIT                                |
| 2023–2025 | UROP advisor: Rina Cao, MIT                                   |
| 2023      | UROP advisor: Kyle Guerre, MIT                                |
| 2023      | UROP advisor: Kristine Bridges, MIT                           |
| 2021      | MSRP advisor: Florencia Corbo-Ferreira, U. of Florida         |
| 2021      | UROP advisor: Mateo Pisinger, MIT                             |
| 2021–2022 | UROP advisor: Ryan Conti, Math + Comp Sci, MIT                |
| 2021      | UROP advisor: Jon Rosario, Math + Comp Sci, MIT               |
| 2020      | UROP advisor: Neosha Narayanan, Mat. Sci., MIT                |
| 2020–2023 | UROP advisor: Sarah Wells-Moran, Wellesley College            |
| 2020–2023 | Project advisor: Max Filter, Computer Science, U. of Virginia |
| 2020–2021 | UROP advisor: Jack-William Barotta, Math, MIT                 |
| 2020      | UROP advisor: Gabriela Alvarez, Mech. Eng., MIT               |
| 2020      | UROP advisor: Jordan Ambrosio, Mech. Eng., MIT                |
| 2020      | UROP advisor: Stephanie Baez, Civil Eng., MIT                 |
| 2020      | UROP advisor: Anna Chau, EE + Comp. Sci, MIT                  |
| 2020      | UROP advisor: Adriana Flores, AeroAstro, MIT                  |
| 2020      | UROP advisor: Meriah Gannon, Env. Eng., MIT                   |
| 2020      | UROP advisor: Adelynn Paik, Sys. Eng., MIT                    |
| 2020      | UROP advisor: Kristopher Vu, Mech. Eng. + Comp. Sci, MIT      |
| 2020      | UROP advisor: Joyce Yoon, Mech. Eng., MIT                     |
| 2015      | Project advisor: Benjamin Lauer, Université de Lorraine       |
| 2015      | Project advisor: Christine Rains, DEVELOP-JPL                 |
| 2015      | Project advisor: Jerry Heo, DEVELOP-JPL                       |
| 2015      | Project advisor: Erika Higa, DEVELOP-JPL                      |
| 2013      | Project advisor: Sandia Akhtar, Caltech                       |

### Thesis Committee Membership

(in addition to mentorship)

#### Internal thesis committees

|           |                                     |
|-----------|-------------------------------------|
| 2023–     | Riley Thomas (PhD, MIT-WHOI JP)     |
| 2023–     | Maia Cohen (PhD, MIT-WHOI JP)       |
| 2023–     | Caroline Needell (PhD, MIT-WHOI JP) |
| 2023–2025 | Evan Kramer (PhD, MIT AeroAstro)    |

|           |                                                   |
|-----------|---------------------------------------------------|
| 2023–     | Kristen Ammons (PhD, MIT AeroAstro)               |
| 2022      | Mathilde Wimez (MS 2022, MIT EAPS)                |
| 2021–2023 | Ana Glidden (PhD, MIT EAPS)                       |
| 2021–2025 | Alan Gaul (PhD, MIT-WHOI JP)                      |
| 2021–2023 | Cadence Payne (PhD, MIT AeroAstro)                |
| 2020      | Taylor Safrit (MS 2020, MIT EAPS)                 |
| 2020      | Anuar Togaibekov (MS 2020, MIT EAPS)              |
| 2019      | Dylan Cohen (MS 2019, MIT EAPS)                   |
| 2018–2020 | Jeffery Mei (PhD, MIT-WHOI JP; defended Aug 2020) |
| 2018–2021 | Eric Stansifer (PhD, MIT EAPS; defended Oct 2021) |

### External thesis committees

|           |                                                    |
|-----------|----------------------------------------------------|
| 2022      | Aaron Stubblefield (PhD 2022, Columbia University) |
| 2021      | Cornelius Quigley (PhD 2021, University of Tromsø) |
| 2019      | Martine Espeseth (PhD 2019, University of Tromsø)  |
| 2018–2021 | Juan Pinales (MS 2021, University of Miami)        |
| 2018–2023 | Stephanie Olinger (PhD 2023, Harvard)              |

### Teaching

|             |                   |                                                                                                    |
|-------------|-------------------|----------------------------------------------------------------------------------------------------|
| 2026 Spring | Caltech Ge193     | Glacier Dynamics                                                                                   |
| 2025 Spring | MIT 12.421/12.621 | Physical Principles of Remote Sensing (undergraduate and graduate)                                 |
| 2024 Spring | MIT 12.203/12.503 | Mechanics of Earth (undergraduate and graduate)                                                    |
| 2024 Spring | MIT 12.512        | Field Geophysics Analysis (graduate)                                                               |
| 2024 IAP    | MIT 12.511        | Geophysics Field Camp (graduate)                                                                   |
| 2023 Fall   | MIT 12.202/12.502 | Flow, Deformation, and Fracture in Earth and Other Terrestrial Bodies (undergraduate and graduate) |
| 2023 Fall   | MIT 12.214/12.507 | Essentials of Applied Geophysics (undergraduate and graduate)                                      |
| 2022 Fall   | MIT 12.421/12.621 | Physical Principles of Remote Sensing (undergraduate and graduate)                                 |
| 2022 Spring | MIT 12.203/12.503 | Mechanics of Earth (undergraduate and graduate)                                                    |
| 2022 IAP    | MIT 12.511        | Geophysics Field Camp (graduate)                                                                   |
| 2021 Fall   | MIT 12.S595       | Geophysics Field Course (graduate)                                                                 |
| 2021 Fall   | MIT 12.202/12.502 | Flow, Deformation, and Fracture in Earth and Other Terrestrial Bodies (undergraduate and graduate) |
| 2021 Spring | MIT 12.S595       | Antarctic Ice Sheet in a Changing Climate (graduate)                                               |
| 2020 Fall   | MIT 12.421/12.621 | Physical Principles of Remote Sensing (undergraduate and graduate)                                 |
| 2020 Spring | MIT 12.203/12.503 | Mechanics of Earth (undergraduate and graduate)                                                    |
| 2019 Fall   | MIT 12.202/12.502 | Flow, Deformation, and Fracture in Earth and Other Terrestrial Bodies (undergraduate and graduate) |

|             |                   |                                                                                                                            |
|-------------|-------------------|----------------------------------------------------------------------------------------------------------------------------|
| 2019 Spring | MIT 12.005/12.520 | Applications of Continuum Mechanics to Earth, Atmospheric, and Planetary Sciences (undergraduate) / Geodynamics (graduate) |
| 2018 Fall   | MIT 12.421/12.621 | Physical Principles of Remote Sensing (undergraduate and graduate)                                                         |

## Academic Service

(internal and external)

|           |                          |                                                                  |
|-----------|--------------------------|------------------------------------------------------------------|
| 2024–2025 | MIT EAPS                 | Chair, Program in Geophysics                                     |
| 2024      | MIT EAPS                 | Department Head search committee                                 |
| 2024      | MIT EAPS                 | Graduate admissions committee                                    |
| 2023      | MIT EAPS                 | Renovation committee (chair)                                     |
| 2023–2025 | MIT-WHOI                 | MIT-WHOI Joint Program Committee                                 |
| 2022–2025 | MIT                      | Warrior Scholar Project                                          |
| 2022–2023 | MIT EAPS                 | Department website committee                                     |
| 2021      | UNAVCO                   | Guest lecturer for InSAR/ISCE workshop                           |
| 2021–2024 | UNAVCO                   | WInSAR Executive Committee member                                |
| 2021–2022 | MIT EAPS                 | Department Lecture Series committee                              |
| 2020–2024 | MIT                      | Common Ground for Computing Education Committee                  |
| 2020      | NASA                     | Surface Deformation and Change – Cryosphere Working Group        |
| 2019–2025 | AGU Journals             | Associate Editor, Journal of Geophysical Research: Earth Surface |
| 2019–2020 | MIT EAPS                 | Task Force 2023 committee member                                 |
| 2018–2025 | MIT EAPS                 | Communications committee member                                  |
| 2018–2021 | Boston Museum of Science | External advisory committee member                               |
| 2018–2019 | MIT EAPS                 | Faculty search committee member                                  |

## Professional Activities

*Chief Scientist and Co-Founder:* Arête Glacier Initiative (2024–)

*Associate Editor:* Journal of Geophysical Research: Earth Surface (2019–2025)

*Scientific Editor:* Annals of Glaciology (2017–2018)

*Member:* International Glaciological Society, International Association of Cryospheric Sciences, Association of Polar Early Career Scientists, American Geophysical Union, European Geosciences Union, American Association for the Advancement of Science

*Notable external committees:* Boston Museum of Science advisory board member; NASA Surface Deformation and Change – Cryosphere Working Group; UNAVCO WInSAR Executive Committee member

## Selected Outreach Efforts

- Faculty Advisor, MIT Student Veterans Association
- Speaker and Lecturer, Warrior Scholar Project

- Speaker, Lowell Observatory public science event: I Heart Pluto Festival
- Military veteran advising and speaking
- MIT Museum: Teen Science Café
- Boston Museum of Science *Pulsar* science podcast
- Gardner Pilot Academy, Allston, MA: Introducing 8th grade math students to glaciers and glaciology
- *Arctic Adventure: Exploring with Technology* at the Boston Museum of Science
- NASA Climate Day presenter
- US Embassy Iceland outreach on Arctic climate
- Iridescent Learning: Teaching children about scientific concepts

## Military Service

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United States Marine Corps

Active duty August 1996–February 2004

Rank: Staff Sergeant

Units: HMX-1, HMM-461, HMM-264, 26th MEU (aboard USS Iwo Jima LHD-7)

Major campaigns: Operation Iraqi Freedom, Combined Joint Task Force–Horn of Africa, Joint Task Force Liberia